

RAMANJEET SAINI

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TECHNICAL SKILLS

Languages: Python (5+ yrs, expert), SQL, Scala, Go, Bash | **Data Engineering:** Apache Spark (PySpark / Scala), Kafka, Flink, dbt, Databricks, Airflow, Argo Workflows, Fivetran, Airbyte, AWS Glue, AWS EMR, Delta Lake | **Streaming:** Apache Kafka, AWS Kinesis, Flink, exactly-once semantics, watermark strategies, event-driven architectures | **Data Warehouses & Stores:** Snowflake, BigQuery, Delta Lake, Databricks Lakehouse, PostgreSQL, DynamoDB, Redis, OpenSearch, Oracle, MongoDB | **Data Quality:** dbt unit tests, Great Expectations, Evidently AI, DataDog, Prometheus, data lineage, SLA / SLI monitoring | **ML & MLOps:** AWS SageMaker, Kubeflow Pipelines, MLflow, PyTorch, XGBoost, LangChain, LlamaIndex, AWS Bedrock | **Cloud & DevOps:** AWS (SageMaker, Bedrock, Lambda, EKS, EMR, Kinesis, Textract, KMS, S3), Kubernetes, Docker, Terraform, Helm, GitLab CI/CD, GitHub Actions

EDUCATION

The University of Texas at Austin May 2023
M.S., Information Technology & Management | Concentrations: Machine Learning, Data Science | GPA: 3.96 / 4.0

Pandit Deendayal Energy University May 2018
B.Tech., Mechanical Engineering

EXPERIENCE

Western Union Austin, TX
Machine Learning Engineer (Data & ML Platform) Aug 2023 – Present

- Modern Analytics ELT Platform:** Improved data freshness from hours to **20 minutes** and cut pipeline recompute by **45%** by re-architecting onto **dbt on Snowflake** with Fivetran / Airbyte ingestion and Airflow orchestration; enforced SLA/SLI with DataDog / Prometheus; eliminated 18 redundant jobs.
- Real-Time ML Feature Platform:** Scaled feature serving to **500M+ reads/day** by designing end-to-end feature infrastructure: **Kafka** for real-time events, **Snowflake** for batch tables, and **Redis** as online feature store at sub-10 ms latency.
- Partner Funding Demand Forecasting:** Reduced idle float by **\$2.5M** and achieved **99% cash-availability SLA** by owning data pipelines and feature engineering for forecasting; orchestrated training on **SageMaker (Prophet, SARIMAX, DeepAR)**; achieved $MAPE \leq 7\%$.
- RAG Document Ingestion Pipeline:** Enabled **26% reduction in analyst handle time** and **19 pp increase in straight-through processing** by building the document embedding pipeline: chunked, embedded, and indexed case documents into **OpenSearch via LlamaIndex** with full PII / GDPR / AML compliance.
- KYC Document Processing Pipeline:** Cut **\$0.9M/quarter** in manual review costs by building the KYC ingestion layer: **AWS Textract + Lambda** for document processing, **DynamoDB** as operational store, and **S3** as document archive with full audit trail.
- ML Model Data Pipelines (Churn & CLV):** Cut model release cycle by **30%** and lifted customer retention by **12 pp** by building automated retraining pipelines on **SageMaker + Kubeflow** with MLflow tracking and Evidently AI drift monitoring.

HEB Digital Austin, TX
Software Engineer — Capstone Intern Dec 2022 – Apr 2023

- Built a REST API with **FastAPI + SQLAlchemy** on containerized **PostgreSQL / Kubernetes** to automate GitLab vulnerability-scan reporting; integrated Slack webhooks and an ML severity classifier — reduced triage effort by **20%** and acknowledgment time by **33%**.

Infosys Limited Pune, India
Data Engineer Jul 2020 – Jun 2022

- Cut infrastructure costs by **34%** by owning large-scale **Apache Spark / AWS EMR** batch ETL/ELT pipelines (500M+ records/week) with Airflow orchestration, ML-based anomaly detection, and **AWS Kinesis** real-time streaming; optimized via cluster right-sizing and spot instances.
- Improved throughput by **12%** via **PySpark + Kafka** → **S3** → **DynamoDB** pipelines; achieved **\$4.7K/month** savings by optimizing **AWS Lambda** functions; led Snowflake / S3 POC improving query performance by **25%**.

Infosys Limited Pune, India
Cloud Engineer Oct 2018 – Jul 2020

- Reduced DBaaS (SQL Server, Oracle, PostgreSQL) downtime by **28%** via CloudWatch alerts and capacity planning; deployed Tableau connectors via GitLab CI/CD with Python data-quality checks — improved reliability by **18%** and operational efficiency by **15%**.

PROJECTS

Flight Deal Finder: Real-Time Cash vs. Miles Comparison Pipeline — github.com/ramanjeetsaini 2025

- Built a personal travel tool that ingests live flight fares and airline award prices via **AWS Lambda** → **S3** + **DynamoDB**; scheduled data refresh and quality checks via **Airflow**; embedded normalized fare data into **FAISS** (<50 ms vector search) for semantic querying; integrated **LangChain + GPT-4o** to rank options by cash-vs-miles value — achieved >0.85 RAGAS answer-relevance score on retrieval quality.

Personal Music Streaming Analytics Dashboard (Kafka + Spark + Airflow) — github.com/ramanjeetsaini 2024

- Built a personal dashboard to analyze own Spotify listening habits; ingested play events in real time via **Kafka** → **S3**; computed batch features (top artists, genre trends, skip patterns) in **Apache Spark**; served aggregates from a **Redis** online store at sub-10 ms; orchestrated daily feature refresh and data quality checks via **Airflow**; visualized insights in a Streamlit app.